Cardiac Rhythms

Professional Development Department



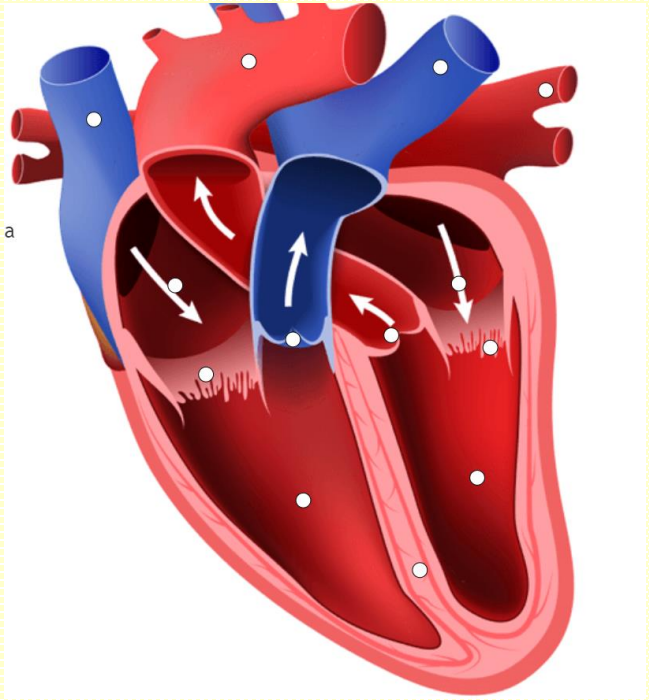
Characteristics of the Heart

Automaticity

Ability of cardiac cells to generate or initiate their own electrical impulse

Excitability

Ability of cardiac cells to Story respond to an electrical impulse



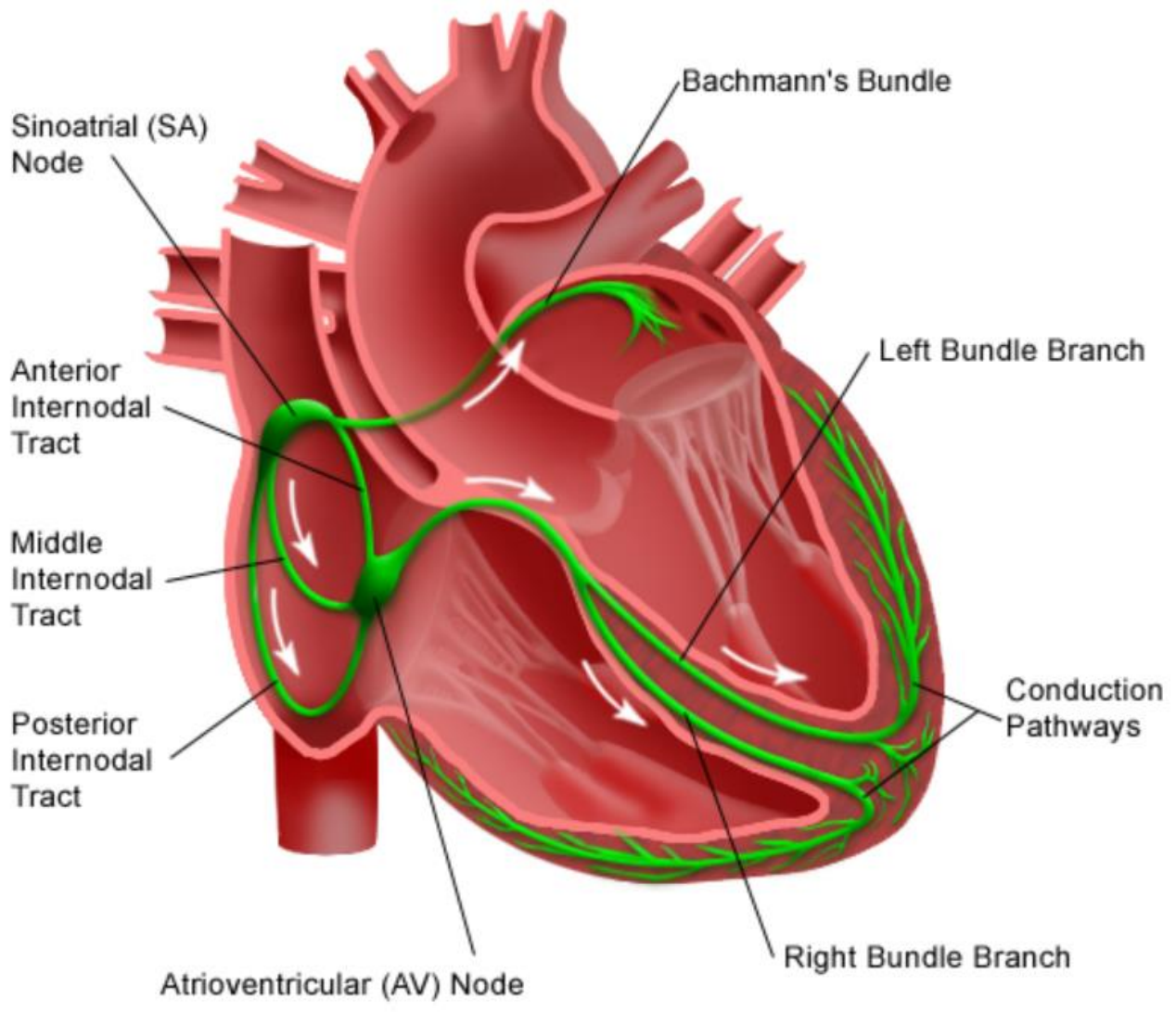
Conductivity

Ability of cardiac cells to generate or initiate their own electrical impulse

Contractility

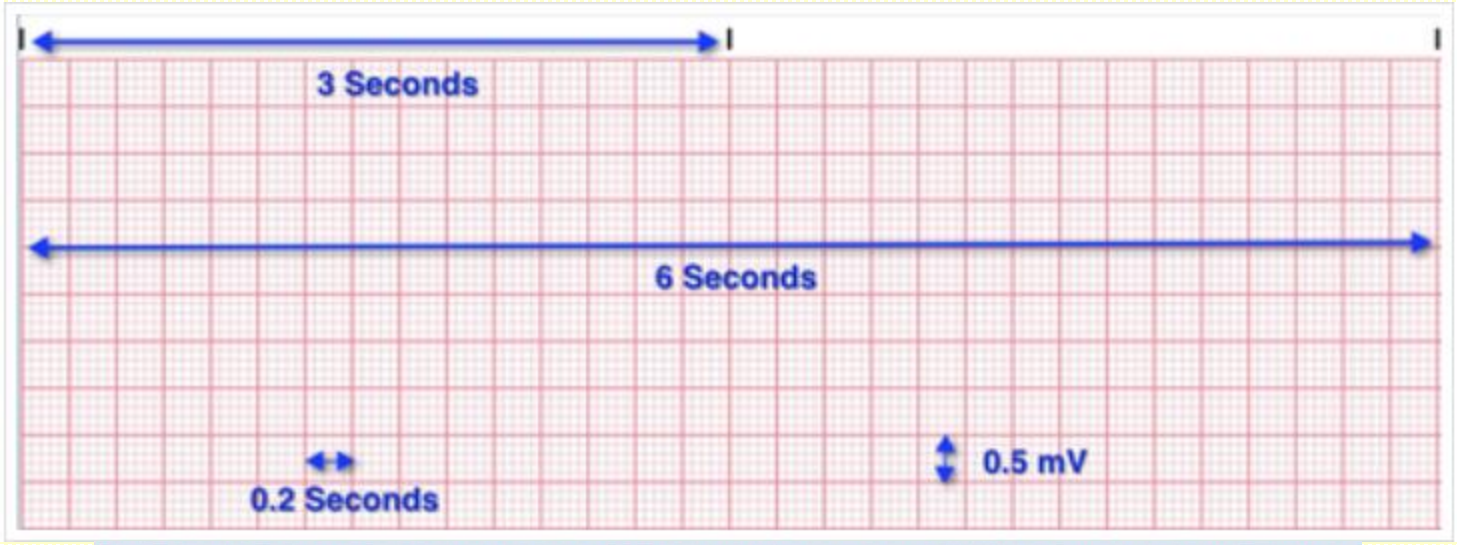
Ability of cardiac cells to shorten causing cardiac muscle contraction in response to an electrical impulse

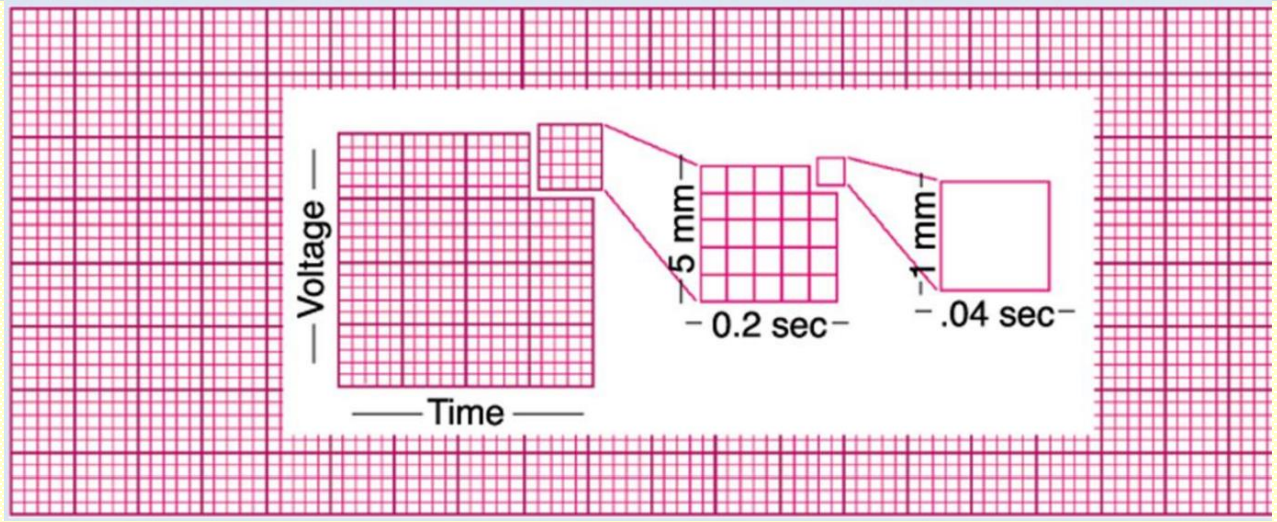
Electrical System of the Heart



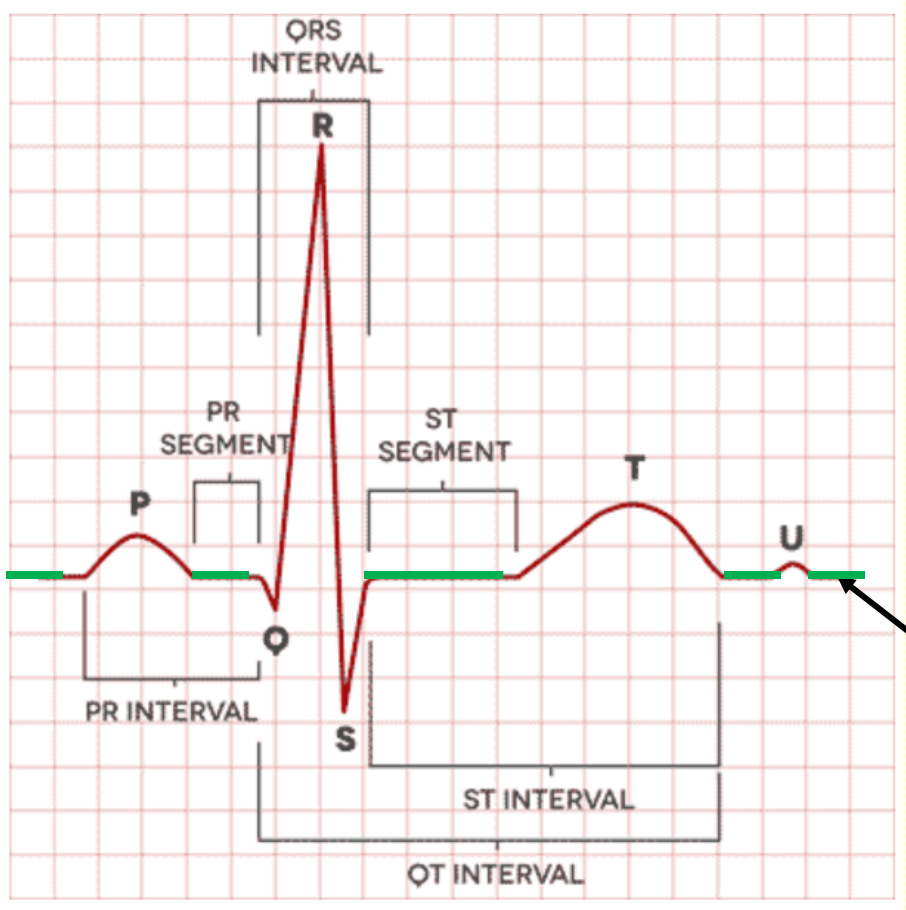
- SA Node " Hearts natural pacemaker"- Rate of 60-100
- AV Junction- Rate of 40-59
- Bundle of His- Rate of 10-39
- Bundle Branches –Rate 10-39
- Purkinje Fibers- Rate of 10-39

EKG Paper


3 Seconds
6 Seconds
0.2 Seconds
0.5 mV


Voltage
Time
5 mm
0.2 sec
1 mm
.04 sec

Cardiac Rhythm Analysis

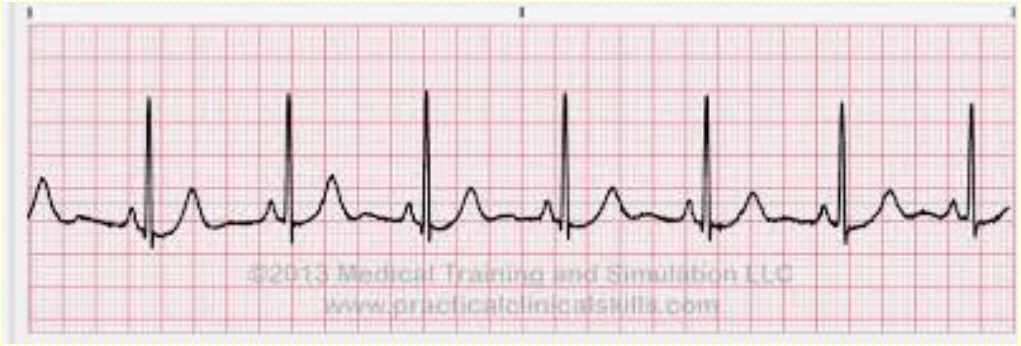


Isoelectric Line or
Baseline

Interpreting a Rhythm Strip

1. ***Heart rate*** – what is the heart rate?
2. ***Rhythm*** – are all R-to-R intervals equal (regular)
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second

Normal Sinus Rhythm

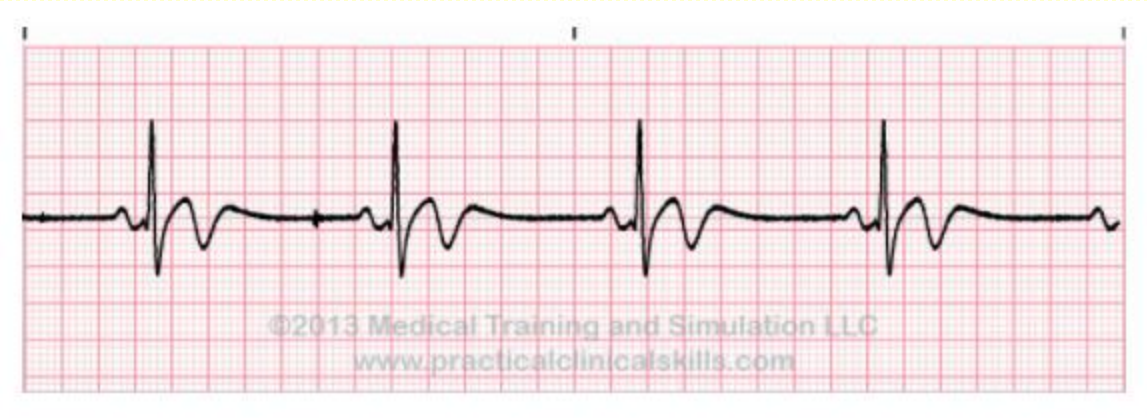


1. ***Heart rate*** – what is the heart rate? **60-100 beats/min**
2. ***Rhythm*** – R to R intervals equal? **Yes**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **Yes**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **Yes**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Sinus Dysrhythmias

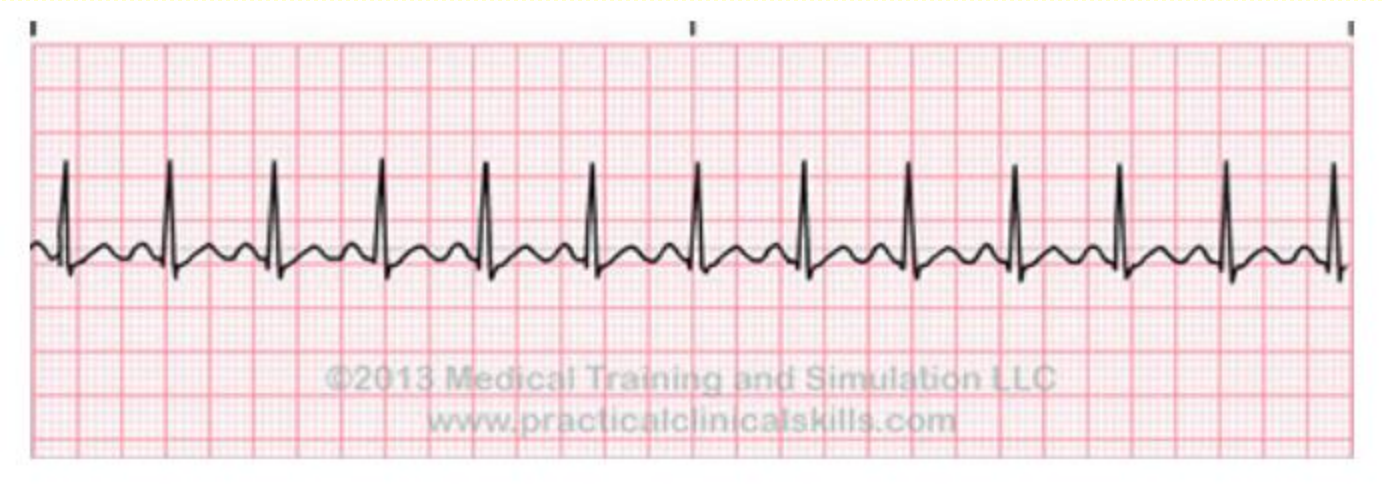
- Sinus Bradycardia
- Sinus Tachycardia
- Sinus Arrhythmia
- Sinus Arrest

Sinus Bradycardia



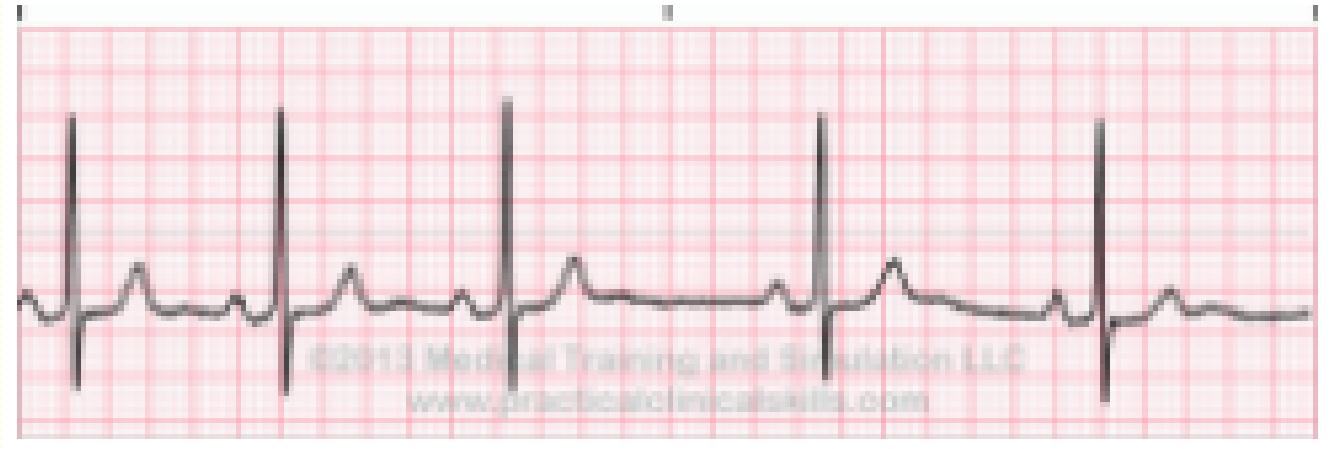
1. ***Heart rate*** – what is the heart rate? **Less than 60 beats/min**
2. ***Rhythm*** – R to R intervals equal? **Yes**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **Yes**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **Yes**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Sinus Tachycardia



1. ***Heart rate*** – what is the heart rate? **Greater than 100 beats/min**
2. ***Rhythm*** – R to R intervals equal? **Yes**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **Yes**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **Yes**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, R to R intervals equal, all within normal range of 0.04-0.12 second **Yes**

Sinus Arrhythmia



1. ***Heart rate*** – what is the heart rate? **Usually 60-100 beats/min**
2. ***Rhythm*** – R-to-R intervals equal? **No. R-to-R intervals shorten with increased HR during inspiration and lengthens with slower HR during expirations**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **No. P-to-P intervals shorten with increased HR during inspiration and lengthens with slower HR during expirations**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **Yes**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second

Sinus Arrest

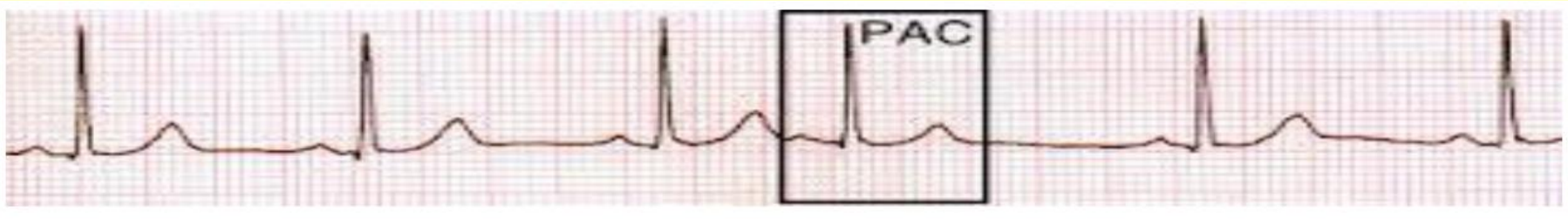


1. ***Heart rate*** – what is the heart rate? **Based on underlying rhythm, could be bradycardic or WNL**
2. ***Rhythm*** – R-to-R intervals equal? **No. R-to-R intervals will be longer for the pause because there is a delay before the SA node generates the next impulse**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **No. P-to-P interval will be longer for the pause because there is a delay before the SA node generates the next impulse**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **Yes**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Atrial Dysrhythmias

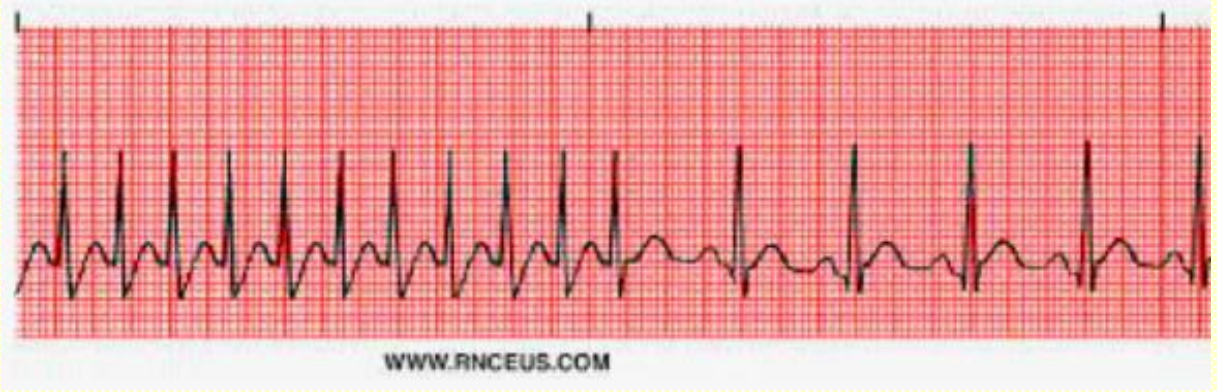
1. Premature Atrial Complex
2. Paroxysmal Atrial Tachycardia/Paroxysmal Supraventricular Tachycardia
3. Supraventricular Tachycardia
4. Atrial Flutter
5. Atrial Fibrillation

Premature Atrial Complex (PAC)



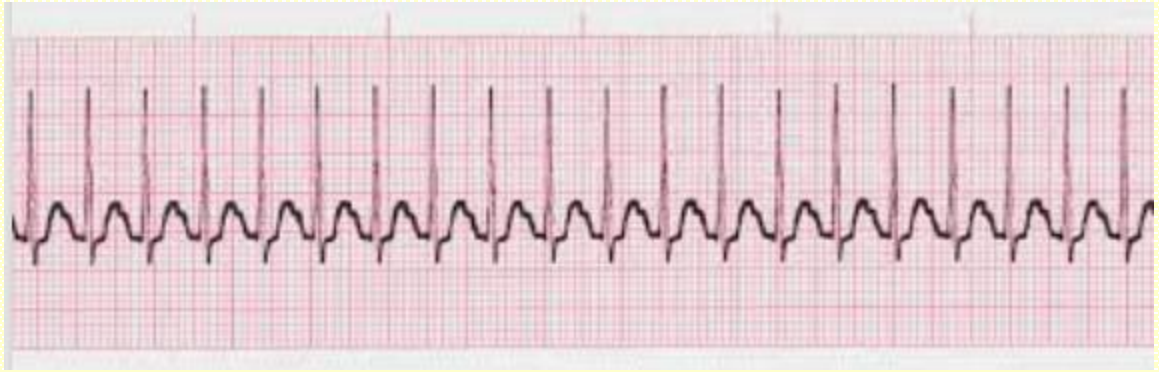
1. ***Heart rate*** – what is the heart rate? **Based on underlying rhythm**
2. ***Rhythm*** – R-to-R intervals equal? **No. R-R interval for the PAC will be less than the previous ones**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **No. The PAC P-wave may appear different in shape or size from the P waves of the underlying rhythm. It may merge with the preceding T-wave (notched appearance) or may be hidden in it (that T-wave will look different from the preceding ones**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **PRI may be different for the PAC**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Paroxysmal Atrial Tachycardia (PAT)/Paroxysmal Supraventricular Tachycardia (PSVT)



1. ***Heart rate*** – what is the heart rate? **Usually greater than 150 beats/min**
2. ***Rhythm*** – R-to-R intervals equal? **Yes**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **No. P-waves are usually hidden**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **Unable to determine because of hidden P-waves**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Unable to identify P waves so unable to determine if QRS complex follows a P wave**

Supraventricular Tachycardia (SVT)



1. ***Heart rate*** – what is the heart rate? **Greater than 150 beats/min**
2. ***Rhythm*** - R to R intervals equal? **Yes**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **No. P-waves are usually hidden**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **Unable to determine because of hidden P-waves**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Unable to identify P waves so unable to determine if QRS complex follows a P wave**

Atrial Flutter



1. ***Heart rate*** – what is the heart rate? **Usually within normal limits**
2. ***Rhythm*** - R-to-R intervals equal? **Yes, if the number of flutter waves is the same before every QRS complex otherwise, it will be different**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **No. Flutter waves "saw-toothed" appearance are produced**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **None exist; P waves do not exist**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Atrial Fibrillation



1. ***Heart rate*** – what is the heart rate? **Ventricular rate may be fast, normal or slow; Atrial rate may be greater that 350 beats/min**
2. ***Rhythm*** – R-to-R intervals equal? **No. R-to-R intervals are irregular**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **No. Erratic fibrillatory waves exist**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **None exists; P waves do not exist**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Junctional Dysrhythmias

1. Premature Junctional Complex (PJC)
2. Junctional Rhythm (a.k.a. Junctional Escape Rhythm) **[HR 40-60 beats/min]**
3. Junctional Bradycardia **[HR < 40 beats/min]**
4. Accelerated Junctional Rhythm **[HR 61-100 beats/min]**
5. Junctional Tachycardia **[HR > 100 beats/min]**

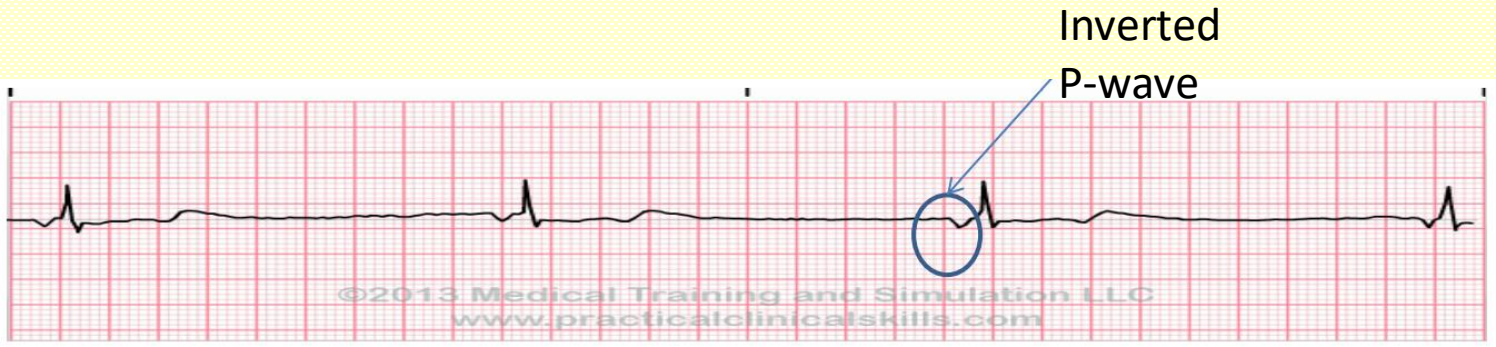
Premature Junctional Complex



Inverted P-wave

1. ***Heart rate*** – what is the heart rate? **Based on underlying rhythm**
2. ***Rhythm –*** R-to-R intervals equal? **No. R-R interval for the PAC will be less than the previous ones**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **No. The PAC P-wave may appear different in shape or size from the P waves of the underlying rhythm. It may merge with the preceding T-wave (notched appearance) or may be hidden in it (that T-wave will look different from the preceding ones**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **PRI may be different for the PAC**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Junctional Escape Rhythm



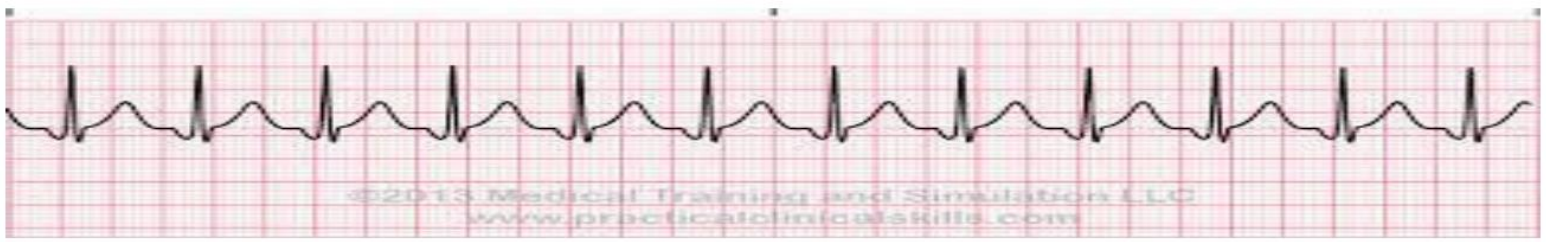
1. ***Heart rate*** – what is the heart rate? **40 -60 beats/min**
2. ***Rhythm –*** R-to-R intervals equal? **Yes**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **P-wave may be inverted, hidden (missing), or after the QRS complex**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **If present (inverted) then usually less than 0.12 second**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Accelerated Junctional Rhythm



1. ***Heart rate*** – what is the heart rate? **61 -100 beats/min**
2. ***Rhythm –*** R-to-R intervals equal? **Yes**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **P-wave may be inverted, hidden (missing), or after the QRS complex**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **If present (inverted) then usually less than 0.12 second**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Junctional Tachycardia



1. ***Heart rate*** – what is the heart rate? **Greater than 100 beats/min**
2. ***Rhythm –*** R-to-R intervals equal? **Yes**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **P-wave may be inverted, hidden (missing), or after the QRS complex**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **If present (inverted) then usually less than 0.12 second**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Atrio-Ventricular (AV) Blocks

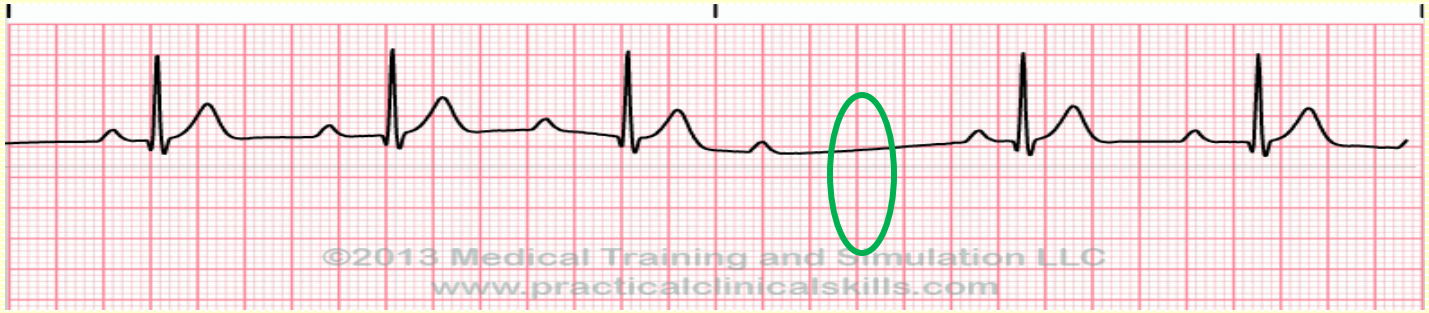
1. First-degree AV Block
2. Second-degree AV Block, Type I (Mobitz I, Wenckebach)
3. Second-degree AV Block, Type II (Mobitz II)
4. Third-degree AV Block (Complete Heart Block)

First-degree AV Block



1. ***Heart rate*** – what is the heart rate? **May be normal, slow, or fast**
2. ***Rhythm –*** R-to-R intervals equal? **Yes**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **Yes**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **Always greater than 0.20 second**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **Yes**

Second-degree AV Block, Type I



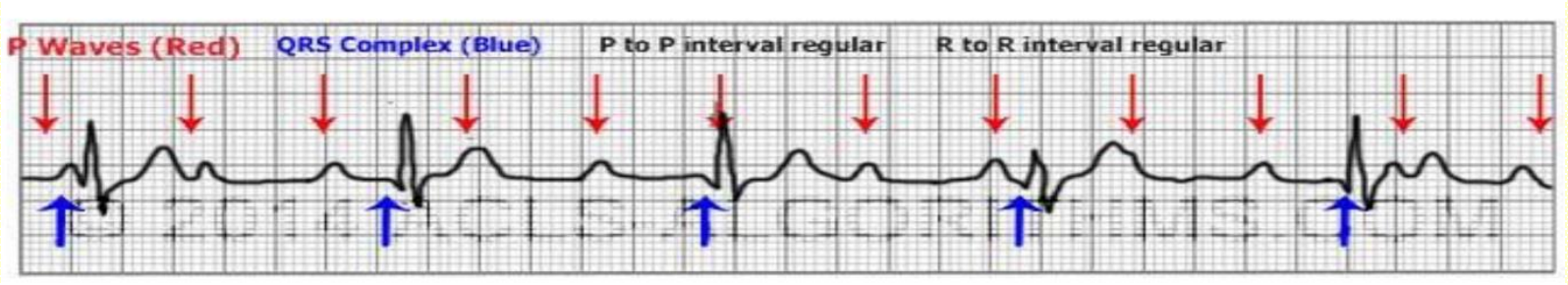
1. ***Heart rate*** – what is the heart rate? **Depends on the number of impulses conducted to the ventricles**
2. ***Rhythm –*** R-to-R intervals equal? **No, the R-to-R remains regular until a QRS complex is dropped (missing)**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **Yes**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **Progressively lengthens until a QRS complex is dropped (missing)**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **No, QRS complex follows every P wave until a QRS complex is dropped (missing)**

Second-degree AV Block, Type II



1. ***Heart rate*** – what is the heart rate? **Depends on the number of impulses conducted to the ventricles**
2. ***Rhythm –*** R-to-R intervals equal? **No because of the dropped (missing) QRS complex**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **Yes**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **All PR intervals are the same until a QRS complex is dropped (missing)**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **No, QRS complex follows every P wave until a QRS complex is dropped (missing)**

Third-degree AV Block



1. ***Heart rate*** – what is the heart rate? **Usually 20-40 second but can vary**
2. ***Rhythm –*** R-to-R intervals equal? **Yes**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **Same size and shape but some may be hidden in the QRS complex or T-wave**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **No. No relationship between P-wave and QRS complex**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **No, QRS complex can be normal size if originates in the AV Junction or wide if originates in the Bundle of His or ventricular tissue, or appears wide and different from the others if the P-wave is hidden in it**

The Heart Block Poem

by the Princeton Surgical Group & nursesLabs

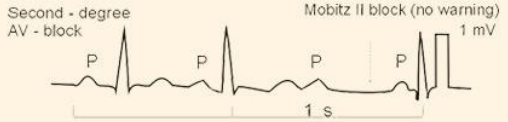
If the **R** is far from **P**,
then you have a **FIRST DEGREE**.



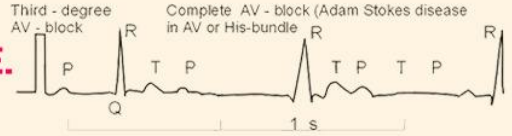
Longer, longer, longer, drop!
Then you have a **WENKEBACH**.



If some **P**s don't get through,
then you have **MOBITZ II.**



If **P**s and **Q**s don't agree,
then you have a **THIRD DEGREE.**



Ventricular Dysrhythmias

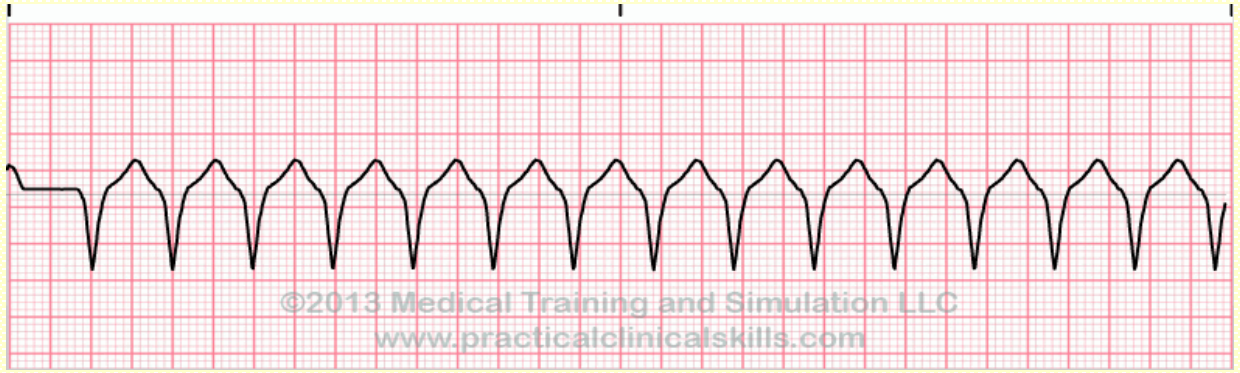
1. Premature Ventricular Complex (PVC)
2. Ventricular Tachycardia (Vtach)
3. Torsades de Pointes
4. Idioventricular Rhythm/Agonal Rhythm
5. Ventricular Fibrillation (Vfib)
6. Asystole

Premature Ventricular Complex (PVC)



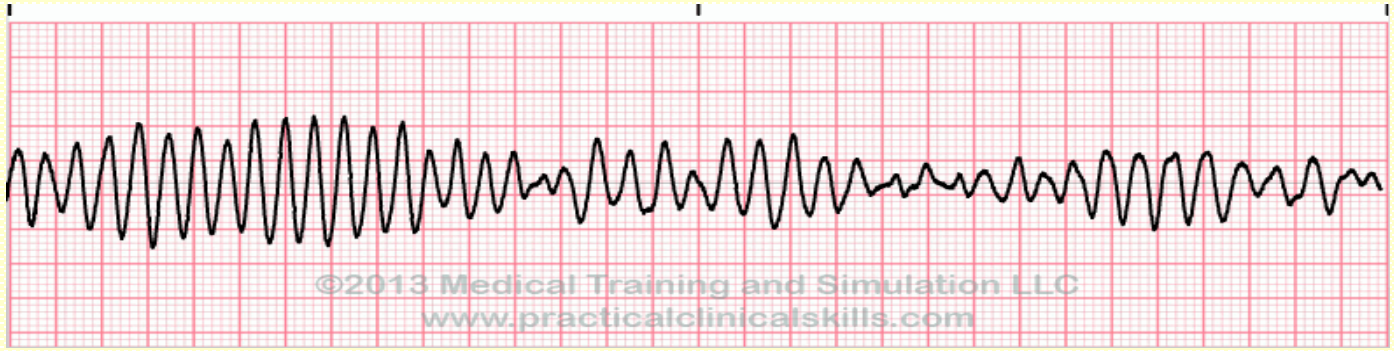
1. ***Heart rate*** – what is the heart rate? **That of the underlying rhythm and the number of PVCs**
2. ***Rhythm –*** R-to-R intervals equal? **Yes, if the underlying rhythm is regular except for the PVC rhythm that comes in early**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **P-waves and intervals are based on the underlying rhythm. A PVC in that rhythm will cause the P-to-P interval to be irregular**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **No. Unable to assess for the PVC**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **No, QRS complex for the PVC is usually wide and bizarre in appearance, is greater than 0.12 second, and may be in the opposite direction of the QRS complexes of the underlying rhythm**

Ventricular Tachycardia (Vtach)



1. ***Heart rate*** – what is the heart rate? **Usually 101- 250 beats/min**
2. ***Rhythm –*** R-to-R intervals equal? **Yes, usually regular**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **P-to-P intervals are not measurable**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **No. PR Intervals are not measurable**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **No, QRS complex is wide and measures greater than 0.12 second**

Torsades de Pointes



1. ***Heart rate*** – what is the heart rate? **Usually 120- 250 beats/min**
2. ***Rhythm –*** R-to-R intervals equal? **No**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **None**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **None**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **No, QRS complex is wide and measures greater than 0.12 second**

Idioventricular Rhythm(Agonal Rhythm)



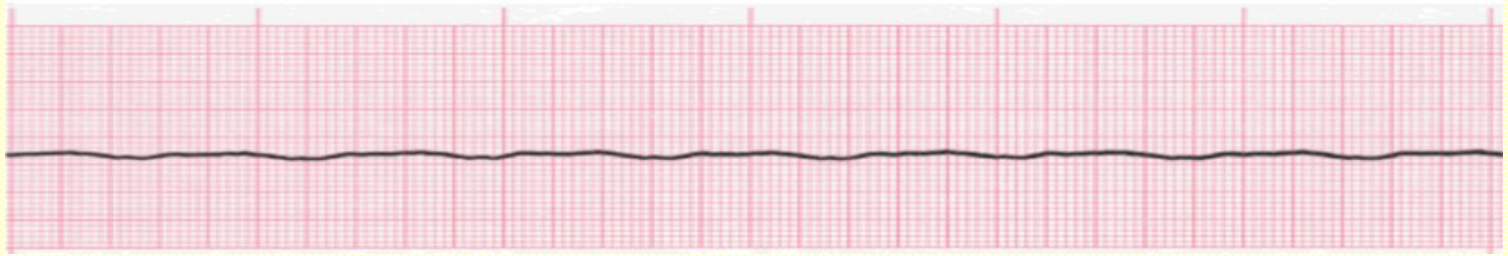
1. ***Heart rate*** – what is the heart rate? **Usually less than 40 beats/min**
2. ***Rhythm –*** R-to-R intervals equal? **May be regular or irregular**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **None**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **None**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **No, QRS complex are wide, bizarre-looking and greater than 0.12 second**

Ventricular Fibrillation (Vfib)



1. ***Heart rate*** – what is the heart rate? **None**
2. ***Rhythm –*** R-to-R intervals equal? **Unable to assess**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **None**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **None**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **No, there are no QRS complexes only a chaotic wavy line is seen**

Asystole



1. ***Heart rate*** – what is the heart rate? **None**
2. ***Rhythm –*** R-to-R intervals equal? **None**
3. ***P-waves*** – present, upright, all look the same, 1 P-wave before every QRS complex, P to P intervals equal **None**
4. ***PR Intervals (PRI)–*** measure each PRI; all present, equal, within normal range of 0.12 to 0.20 second **None**
5. ***QRS complexes*** – examine and measure each QRS complex; present, all look alike, QRS complex after every P wave, all within normal range of 0.04-0.12 second **None**

Additional Topics



Pulseless Electrical Activity (PEA): rhythm may appear normal but the patient has no pulse. Always check your patient's pulse for mechanical capture of the heart i.e., contraction



Paced Rhythm: If the atrium is paced then will see a pacer spike followed by a P-wave. If the ventricle is paced, then the pacer spike is followed by a wide QRS indicating electrical capture has occurred



ST-Elevated MI (Stemi): The ST-segment is above the the isoelectric line (baseline)

Thank you!

Reference

Atwood, S., Stanton, C., Storey-Davenport, J. (2019). Introduction to basic cardiac dysrhythmias (5th ed.). Jones&Barlett Learning: Burlington, MA